

D Opener example with InstructGPT

This quote was obtained through the OpenAI playground for text-davinci-002 on May 15th 2023. The model text-davinci-001 consistently generates better responses for the same prompt. GPT-3 itself (i.e. davinci) mainly gives nonsensical answers. In the following, the prompt is *italic* and the completion **bold**. The completion was generated with a maximum of 10 tokens and a temperature of 0:

User: "Have you seen my phone?"

GPT-3: "Yes, I have seen your phone."

The opener example is used to introduce the problem we are looking at, and not to judge the model used to generate it. In fact, although text-davinci-002 sometimes completes conversations in a way that is unexpected according to pragmatic language usage, it is one of the better models when evaluated few-shot.

E Background on implicature

The first influential consideration of implicature is Grice (1975). In his work, Grice continues the trend of moving away from purely logical accounts of language started by Wittgenstein (1921) by hypothesising implicatures arise in conversation when some mutually agreed upon maxims seem to be violated. For example, if we agree on only making relevant contributions to conversation, Juan's response in the introduction seemingly violates this maxim—after all, he starts talking about work when Esther asks him about a party. However, because Juan agreed to be relevant he must be implying that having to work means he cannot come to the party. Grice contrasts conversational implicatures that arise through context with conventional implicatures. These are implicatures where the *conventional* meaning of the word determines what is implicated. An example given by Grice is the following sentence: “he is an Englishman; he is therefore brave.”. Grice notes that this sentence does not literally state that an Englishman being brave is a direct consequence of him being English, but it's implied by the conventional meaning of the word ‘therefore’.

Since then, issues with the Gricean cooperative principle have been pointed out by many (Levinson, 1983; Sperber and Wilson, 1986; Davis, 1998; Lepore and Stone, 2014). The most influential alternative theory is relevancy theory by Sperber and Wilson (1986). They do away with the cooperative principle and instead theorise implicatures arise because speakers try to produce utterances that are both as relevant as possible and require the least effort to process. Another point of contention is the incorporation of conventional implicatures on the pragmatics side. Bach (1999) argues that there is no such thing as conventional implicatures, and they are simply instances of something else. Based on a thorough treatment of what Grice calls conventional implicatures, Bach argues all examples of it can be filed under other concepts within semantics, like utterance modifiers (called “utterance modifiers” instead of “sentence modifiers” because they go against the semantic content of the rest of the sentence). Potts (2005) also argues that to explain conventional implicatures we can stay on semantic turf. Indeed, even Grice himself says conventional implicatures derive from the meaning of the words, not from conversational context. However, Potts does not claim conventional implicatures do not exist, but instead argues they arise by a combination of lexical meaning and novel ways of combining words—the latter being the well-known principle of compositionality, an important part of semantics, not of pragmatics. Potts provides us with an illuminating demarcation between conventional and conversational implicatures. Conventional implicatures are never negotiable by context, whereas conversational implicatures are context-dependent and can always be cancelled without causing incoherent discourse. Consider again the sentence “he is an Englishman; he is therefore brave.” and the sentence “Eddie has three bicycles” (implicating that Eddie has exactly three bicycles and not more). The former sentence can not be cancelled by new context without contradiction, whereas for the latter, if we continue saying “In fact, Eddie has 10 bicycles, he is a bicycle junkie”, we have cancelled the implicature. This demarcation clearly puts conventional implicatures on the semantic side, and conversational implicatures on the pragmatic side. Potts goes on by providing a formal theory for conventional implicatures.

In later work, Potts (2006) describes how pragmatic pressures interacting with context cause conversational implicature to arise. He shows how sensitive conversational implicatures are to small changes in the context. Novel information about a speaker's belief state might completely change what is implied. There are many more models of implicature that aim to explain how humans understand language in context. Most notably, Frank and Goodman (2012) formalise the view that speakers produce utterances that are helpful and not longer than necessary with a Bayesian model called the rational speech act (RSA). Many variants on the RSA framework have since been proposed. For example, Goodman and Frank (2016) extend it to handle nonliteral uses of language, like irony, and metaphor. In the context of computational models, prior work uses insights from pragmatics to show that the use of certain words can make a language model produce biased completions (Patel and Pavlick (2021), e.g. saying someone “claimed” something rather than “said” something), and inform bias and sentiment classifiers (Greene and Resnik, 2009; Recasens et al., 2013).

In this work, we focus on conversational implicatures and not on conventional implicatures. All conversational implicatures are negotiable by context, but the way they depend on context can be different. Grice (1975) identifies generalised conversational implicatures and particularised conversational implicatures. The former require little or no context to be resolved. For example, “some athletes smoke” can imply “not all athletes smoke”, but might also imply “I do not know whether all athletes smoke” when it is a response to the question “do you know whether all athletes

smoke?” (Davis, 2019). The latter only arise in certain contexts. For example, the response “I have an early morning” to the question “do you want to stay for a drink?”.